

RFI Report for January 2025

Executive summary

Overall, the samples collected during the month of January are minimally contaminated by RFI emissions. However, similar to the finding in the [October 2024 report](#), the dataset obtained on the [17th of January 2025](#) exhibits contaminants with comb-like patterns which mainly affect the auto and short baseline cross correlations, potentially due to noise generating activities at/close to the core. It should also be highlighted that there appears to be an RFI emission at 687.30 MHz, with a fraction of time flagged value of 0.27, in all products of the observation done on the [22nd of January 2025](#).

Looking at the summarized version of the January 2025 RFI occupancy. An unusual trend in RFI over frequency compared to previous months is observed. While the fractional RFI as a function of time remains consistent with past observations, we see significantly higher RFI levels across a broad range of frequency intervals. Notably, some of these affected frequencies were previously considered RFI-free for MeerKAT.

This unexpected increase in RFI as a function of frequency could be attributed to several factors, including new interference sources, changes in telescope hardware, or external environmental factors.

1. Introduction

The purpose of this document is to report the current status of RFI on-site. Given the construction work that has been going on over the last couple of months, it is therefore imperative to monitor the construction related activities on site that could potentially generate RFI signals which, in turn, contaminate the science observations. Preferably these reports are to be released on a weekly basis (or every two weeks).

2. Method

The flags data, specifically *ingest_flags*, are considered in these analyses, as they can be used as proxies for RFI frequency occupancy, i.e. RFI emission as a function of frequency channels. The strategy is to select two to three observations per week and report the status of

the RFI contamination on site based on the flags of each dataset (of each observation). For each observation

- Autocorrelation flags of a few *scans* corresponding to the *gain calibrator* are selected. This is due to the fact that, at the time of writing, there are some issues with accessing a big chunk of an entire dataset.
- Autocorrelation flags, which are three dimensional arrays, i.e. *time: frequency channels: correlation products*, are averaged over both time and correlation products.
- Autocorrelation flags become a one dimensional array which is plotted in order to check the RFI occupancy in the frequency channels.

The cross correlation flags are also investigated in order to see how much the RFI emission affects the correlation products which are the main interest in an interferometric observation. The procedure for the cross correlation flags is the same as that of autocorrelation ones. To further go into more detail, the flags corresponding to short and long baselines are split into two parts, and each part is plotted separately. Based on some datasets that were analyzed previously, the U-band observations are the ones that are most affected by the current RFI emissions on site. Hence, the majority of the datasets that will be inspected is in U-band, nevertheless observations in both L and S bands will be looked at too.

In what follows, it is worth noting that the vertical gray shaded areas in all plots denote the known RFI emissions in U-band.

3. Results

a. 2025, 02 January

The correlation products of the data collected on the 2nd of January 2025 are presented in Figures 1, 2 and 3. Some relatively weak RFI emissions (unknown), with a fraction of the time flagged less than 0.1, are noticeable in the autocorrelation flag, especially in the lower part of the band. The cross correlation products, on the other hand, are relatively clean overall, although a few channels in the short baselines are weakly affected.

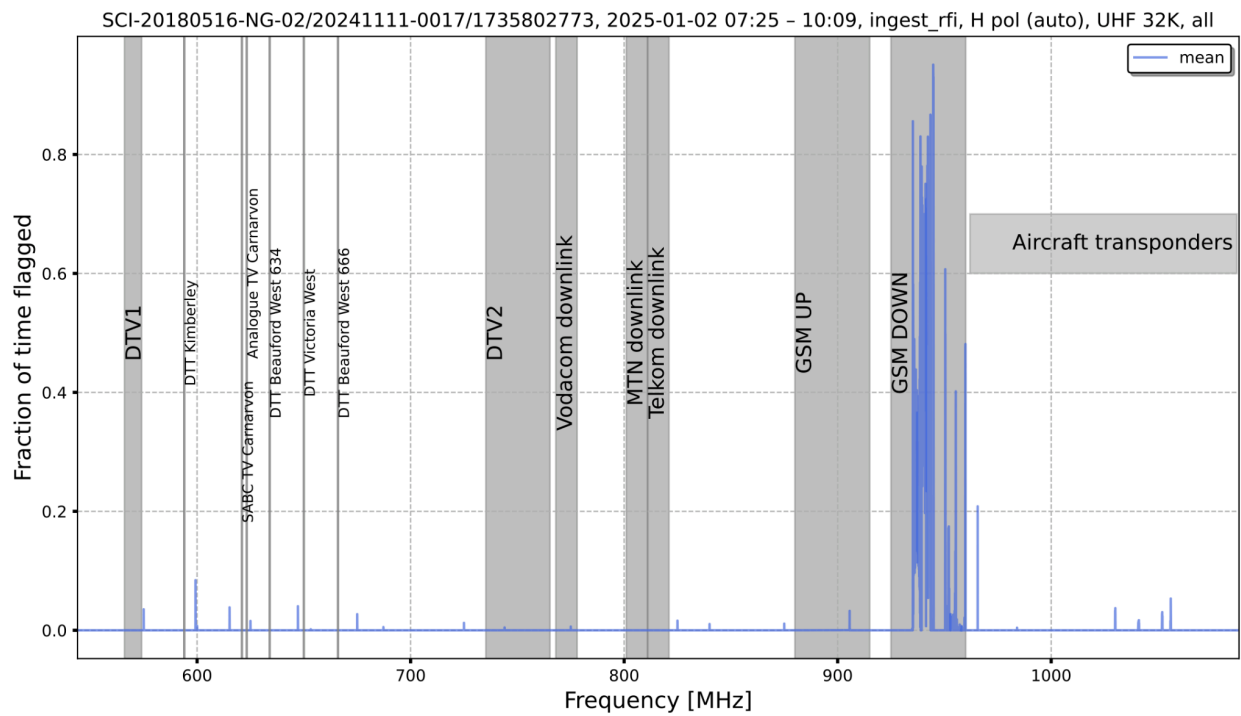


Figure 1: Mean of the autocorrelation flags of SB_ID:20241111-0017 dataset.

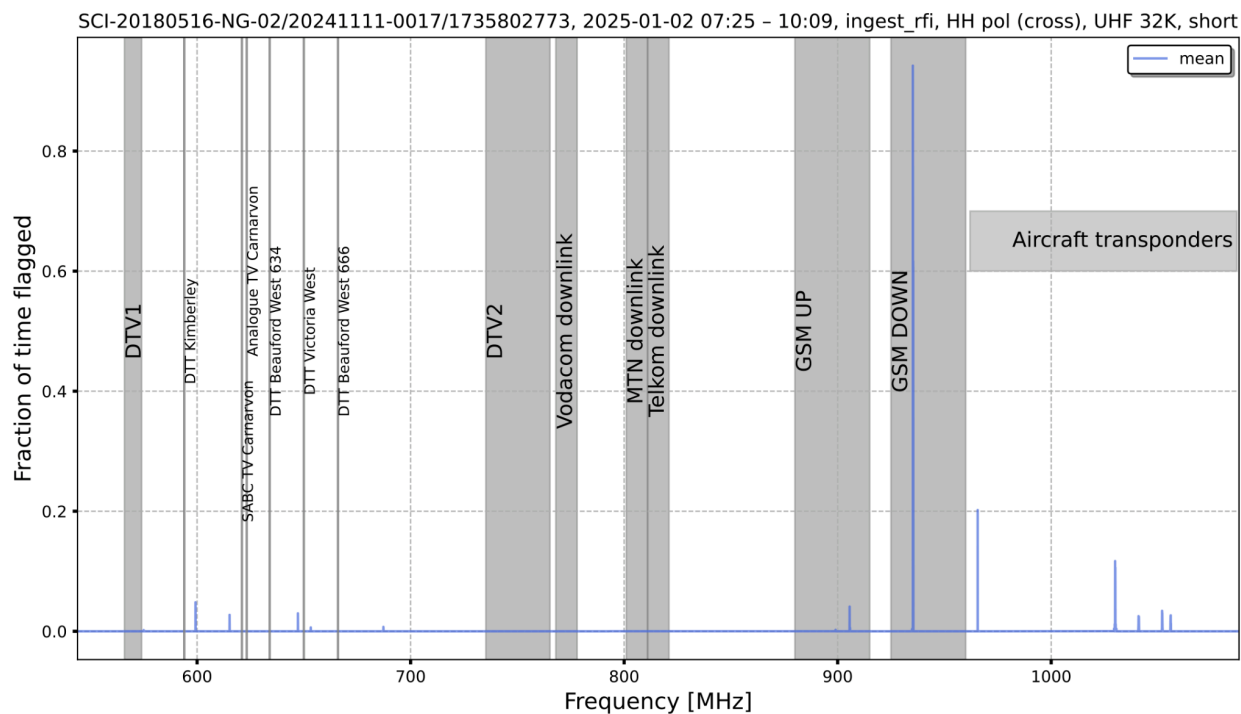


Figure 2: Mean of the cross correlation flags of the short baselines of SB_ID:20241111-0017 dataset.

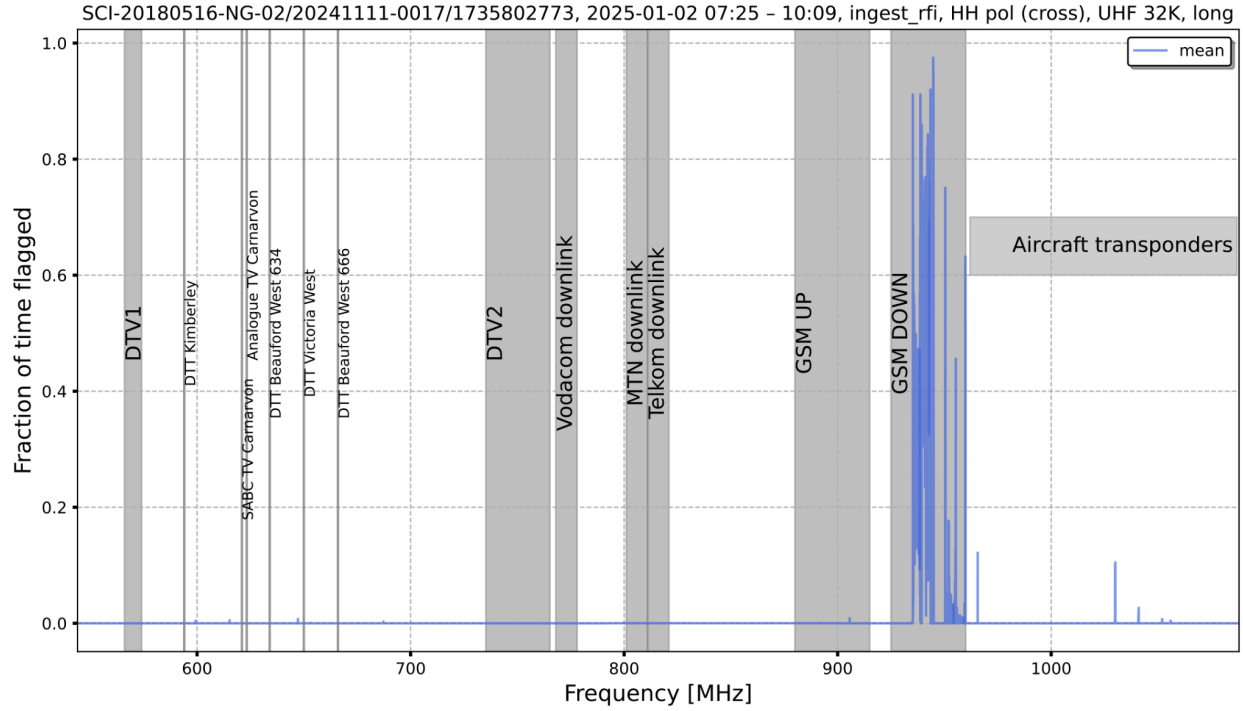


Figure 3: Mean of the cross correlation flags of long baselines of SB_ID:20241111-0017 dataset.

b. 2025, 05 January

The dataset from an observation carried out on the 5th of January 2025 exhibits a few contaminated channels in its autocorrelation product as shown in Figure. 4. The cross correlation flags appear to be clean (see Figures 5, and 6).

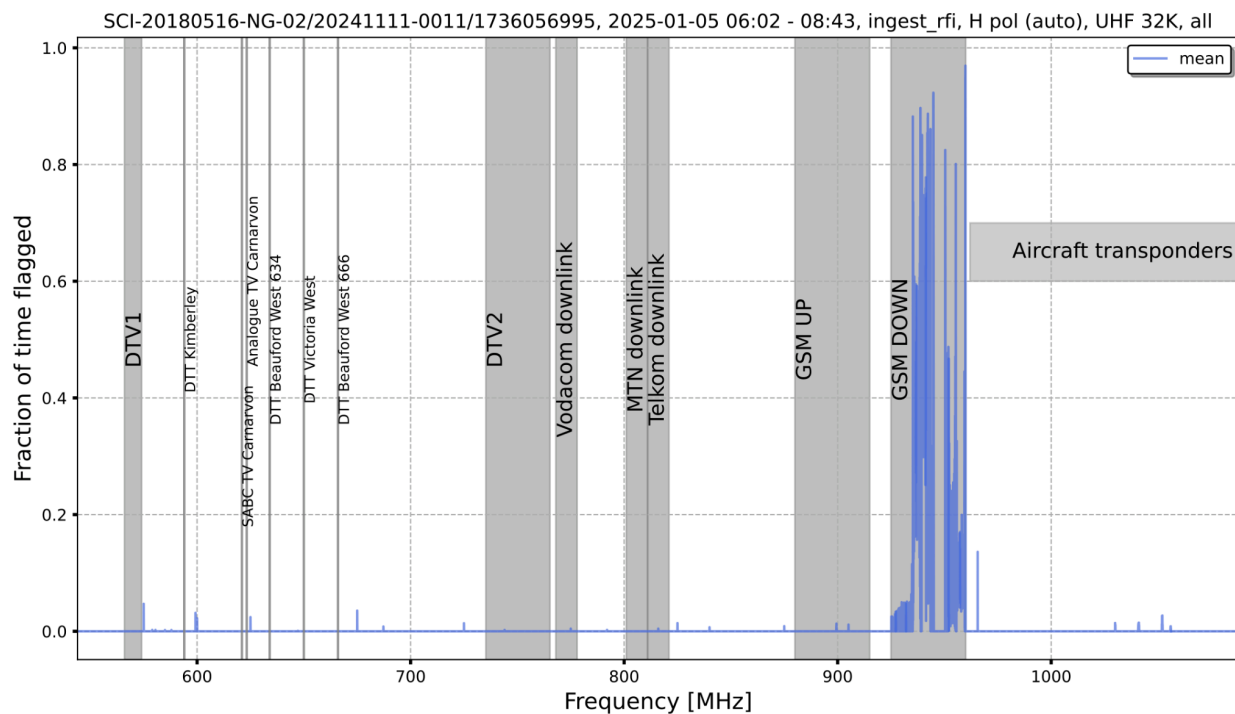


Figure 4: Mean of the autocorrelation flags of SB_ID:20241111-0011 dataset.

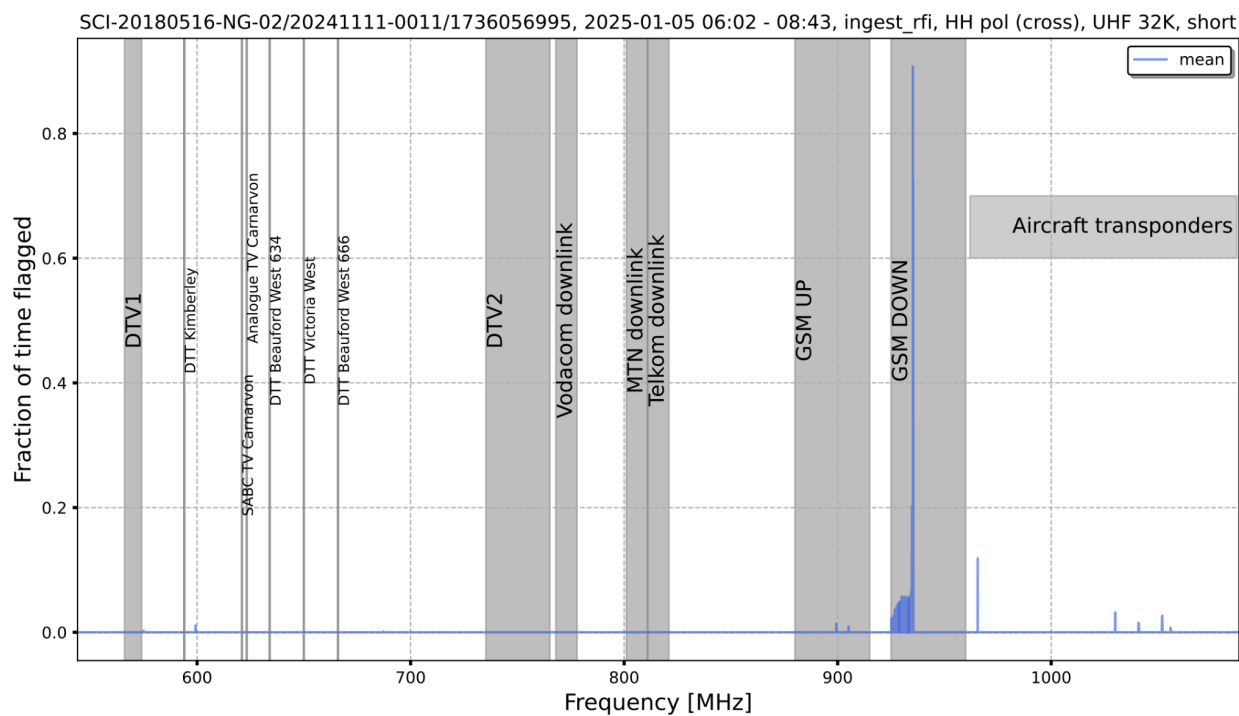


Figure 5: Mean of the cross correlation flags of short baselines of SB_ID:20241111-0011 dataset.

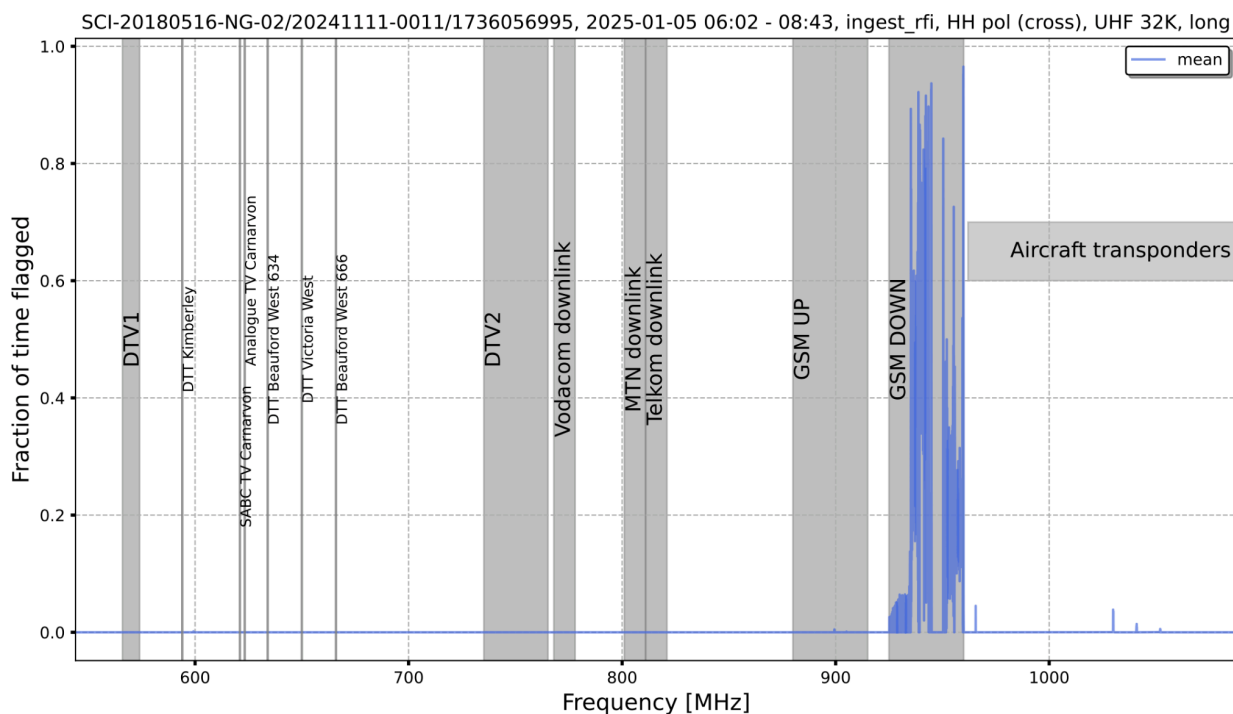


Figure 6: Mean of the cross correlation flags of long baselines of SB_ID:20241111-0011 dataset.

c. 2025, 08 January

The flags plotted in Figures 7, 8, and 9, from an observation on the 8th of January 2025, show a clean bandpass, free of unknown RFI emissions, overall.

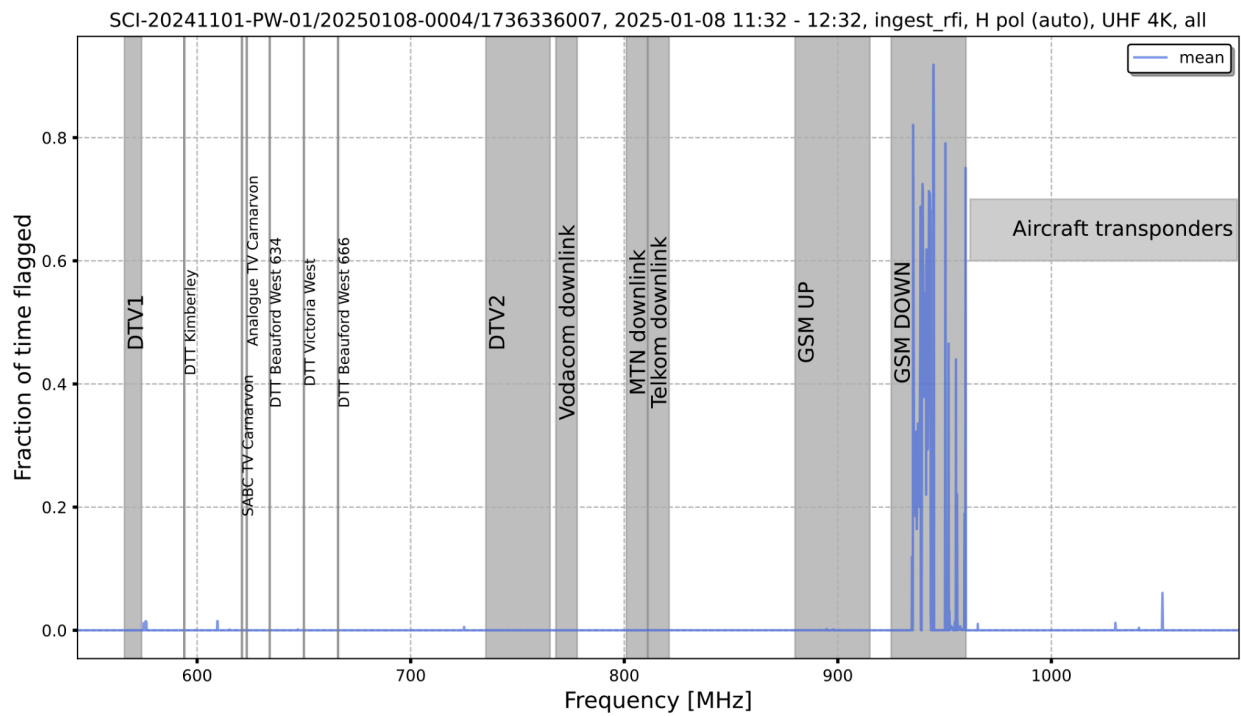


Figure 7: Mean of the autocorrelation flags of long baselines of SB_ID:20250108-0004 dataset.

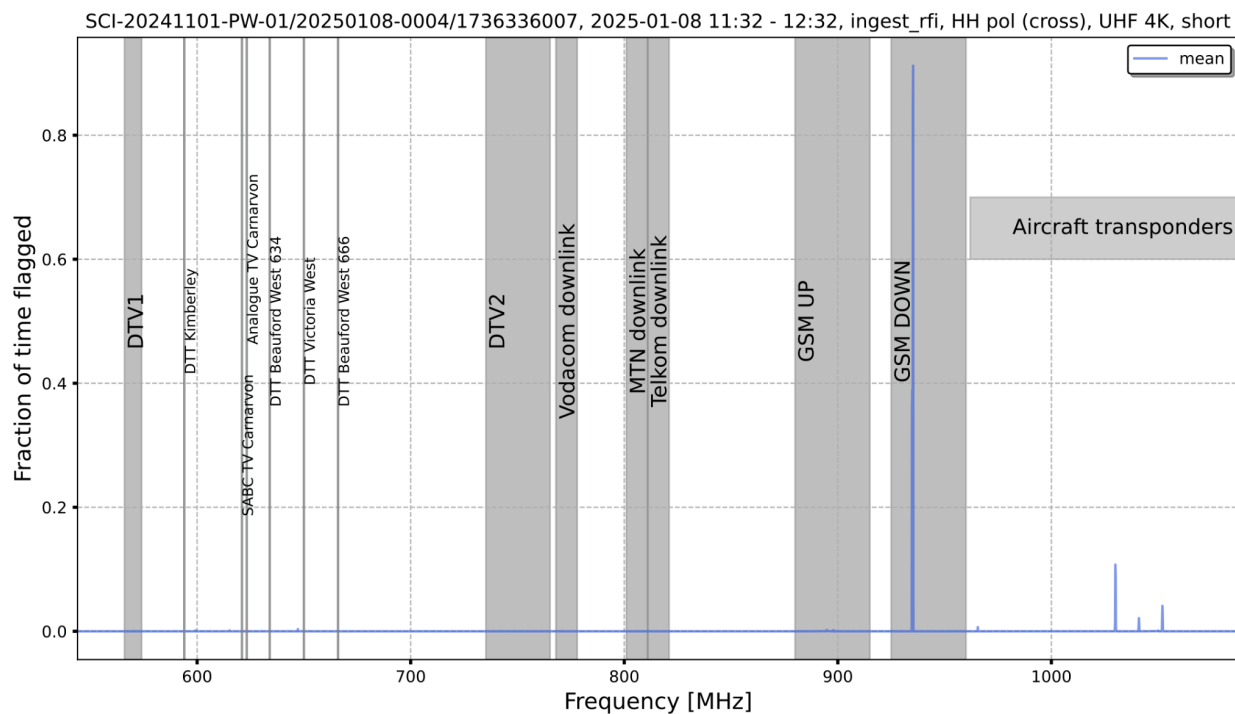


Figure 8: Mean of the cross correlation flags of short baselines of SB_ID:20250108-0004 dataset.

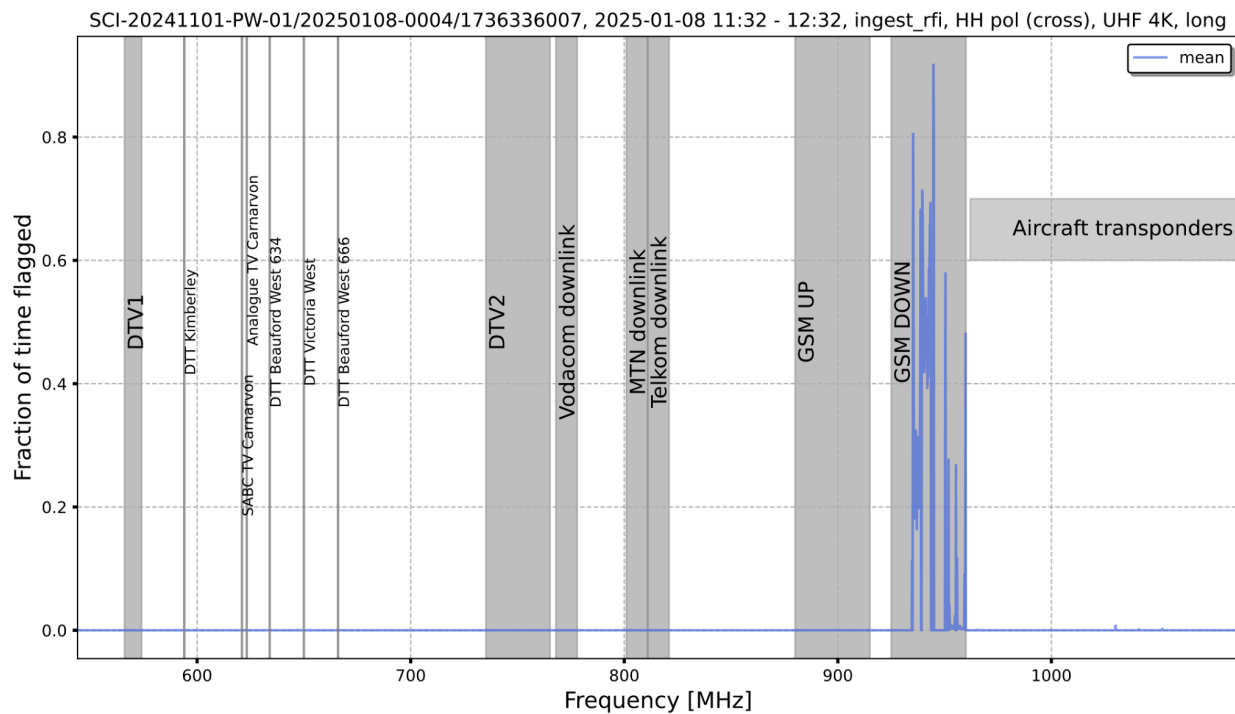


Figure 9: Mean of the cross correlation flags of long baselines of SB_ID:20250108-0004 dataset.

d. 2025, 14 January

Unlike in the two previous sections ([subsections b, c](#)), the autocorrelation flag from this observation conducted on the 14th of January 2025 shows that some channels are weakly contaminated by unknown RFI emissions (see Figure 10), whose fraction of the time flagged is 0.05. There seems to be a channel (around 700 MHz) affected by a weak RFI in the short baselines (see Figure 11), but in general the cross correlation products are relatively clean (see Figures 11 and 12).

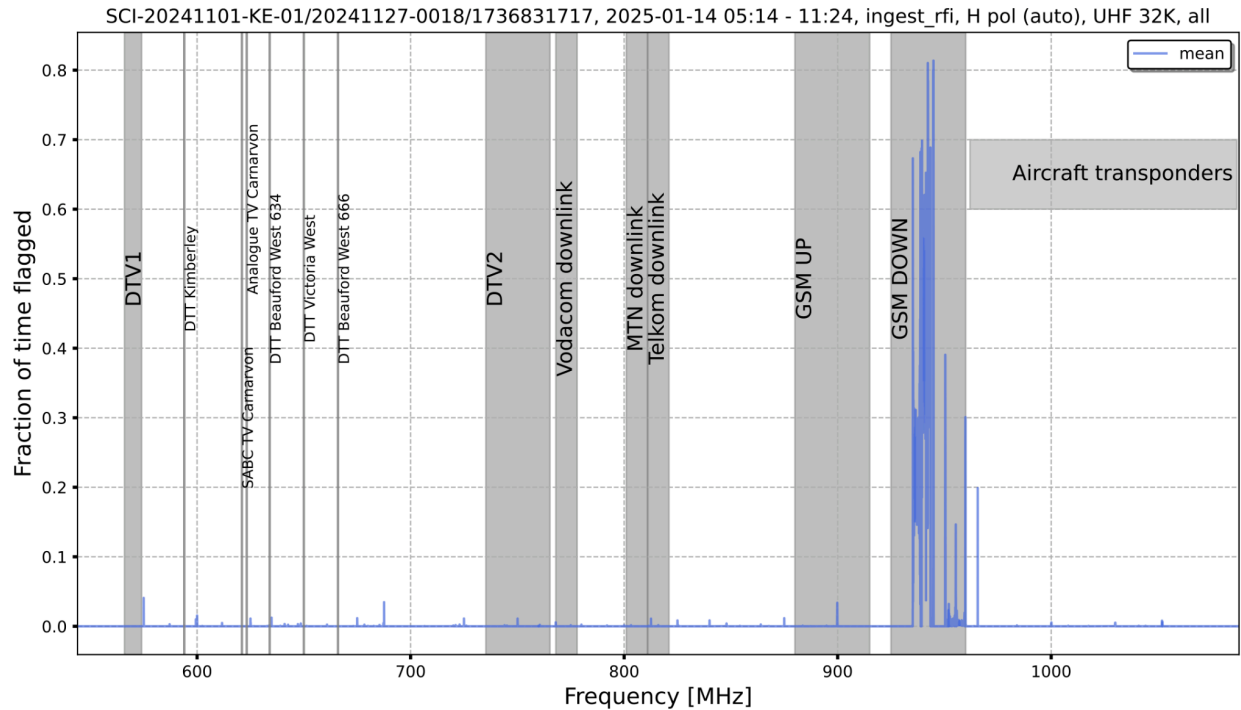


Figure 10: Mean of the autocorrelation flags of SB_ID:20241127-0018 dataset.

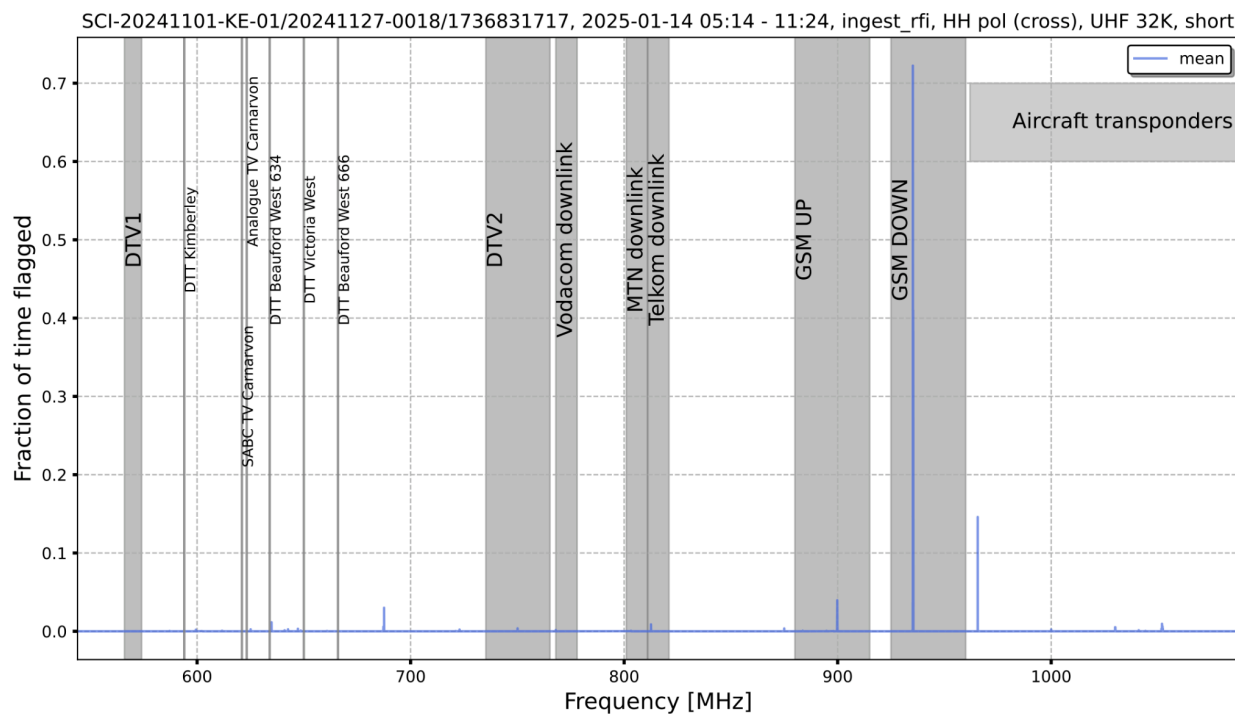


Figure 11: Mean of the cross correlation flags of short baselines of SB_ID:20241127-0018 dataset.

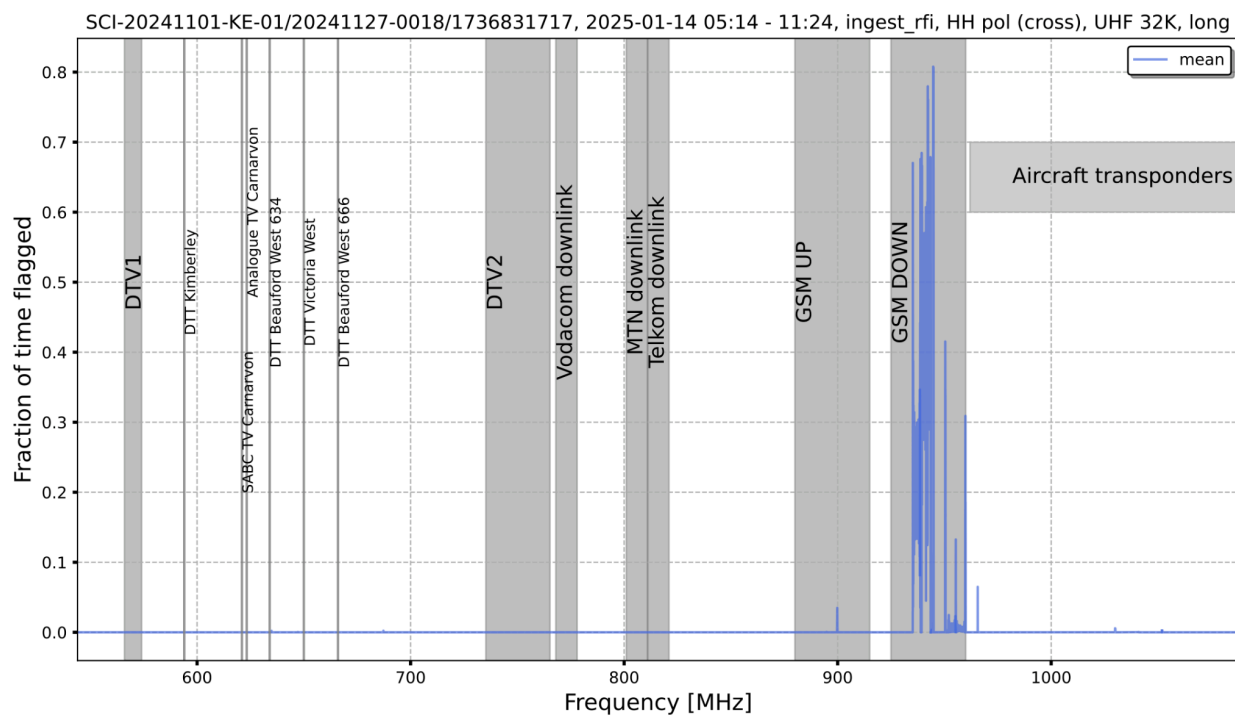


Figure 12: Mean of the cross correlation flags of long baselines of SB_ID:20241127-0018 dataset.

e. 2025, 17 January

The autocorrelation flag of an observation taken on the 17th of January 2025 clearly shows a comb-like RFI contaminating a considerable amount of channels (see Figure 13). Although fewer, some channels still seem to be affected in the short baseline cross correlation as opposed to the long baseline flag which appears to be clean. This indicates that the antennas at/close to the core are those that are mainly contaminated by the RFI emissions.

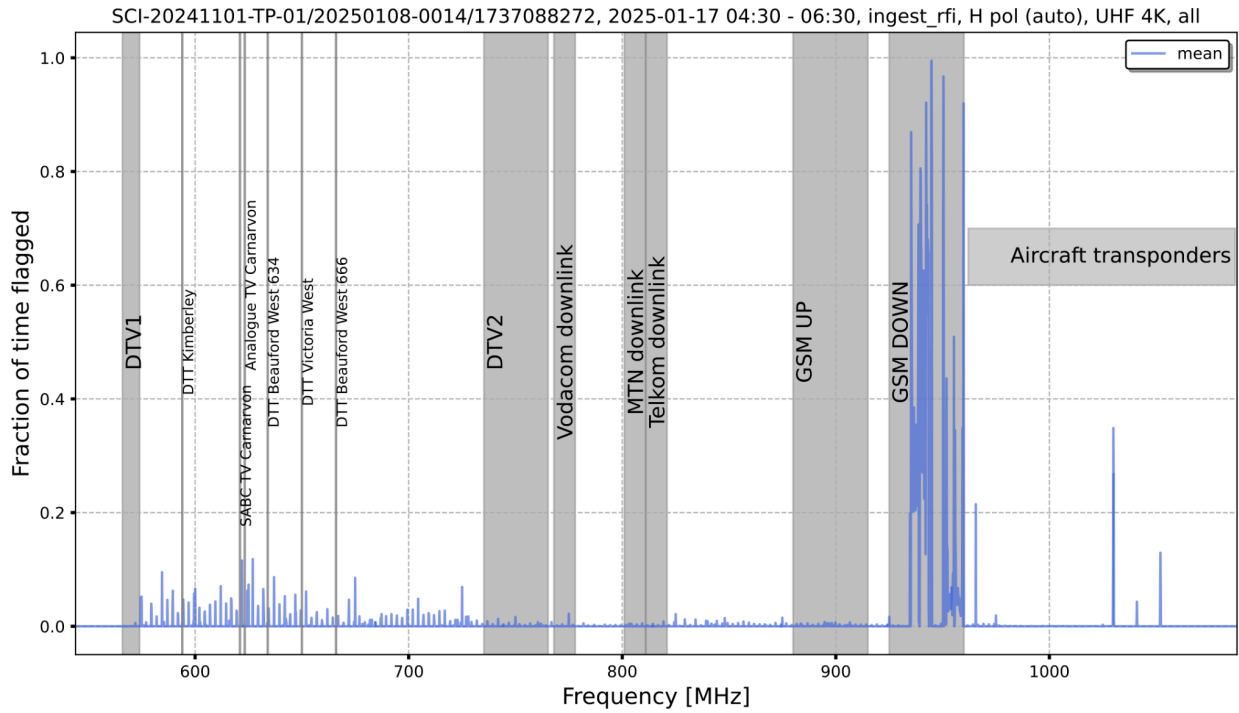


Figure 13: Mean of the autocorrelation flags of SB_ID:20250108-0014 dataset.

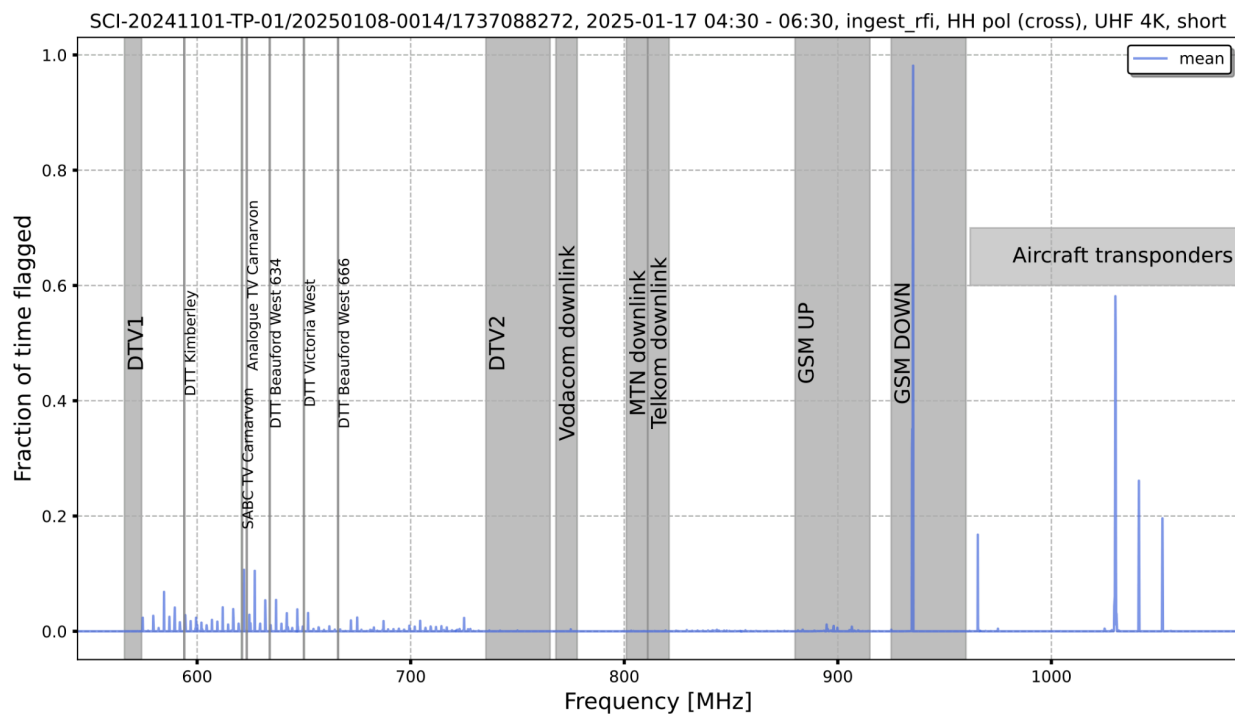


Figure 14: Mean of the cross correlation flags of short baselines of SB_ID:20250108-0014 dataset.

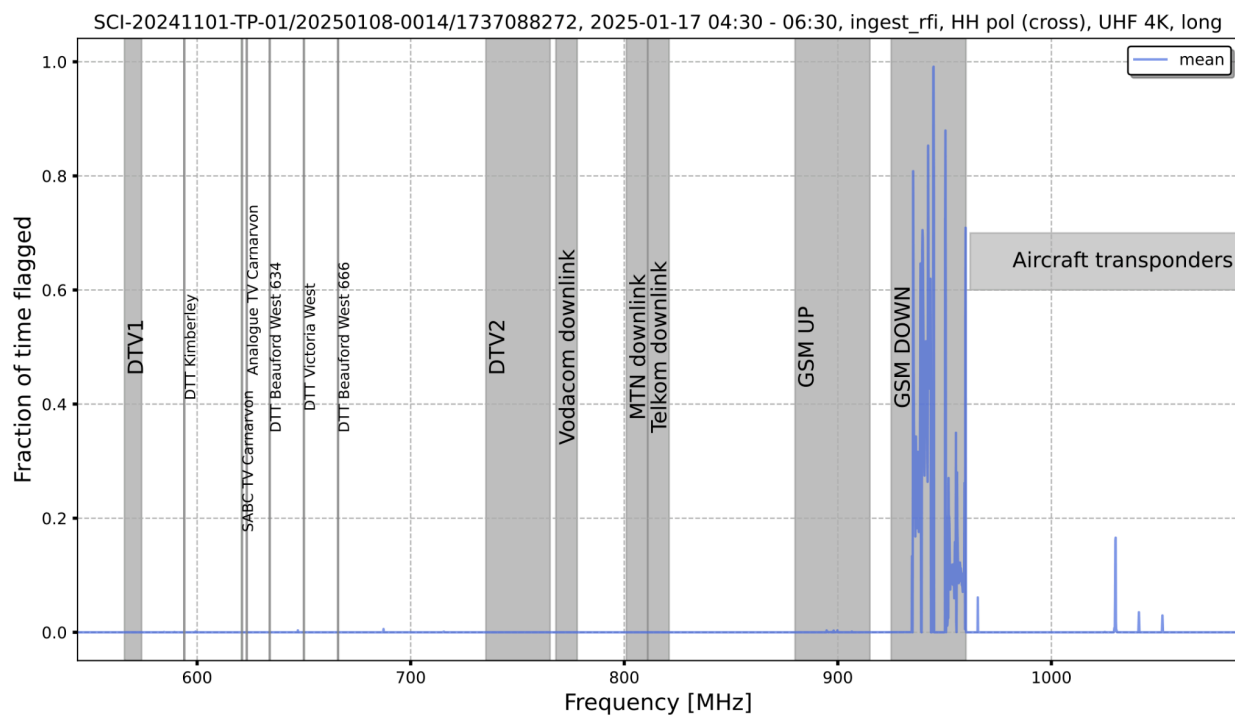


Figure 15: Mean of the cross correlation flags of long baselines of SB_ID:20250108-0014 dataset.

f. 2025, 22 January

Results from the data collected on the 22nd of January 2025 show that there are a few channels affected by RFI emissions in the autocorrelation products (Figure 16), and the cross correlation counterparts (Figures 17 and 18) are clean in general. However, there seems to be a relatively strong RFI, with a fraction of time flagged of about 0.27, at 687.30 MHz in all products. At the time of writing this contaminating noise is not known.

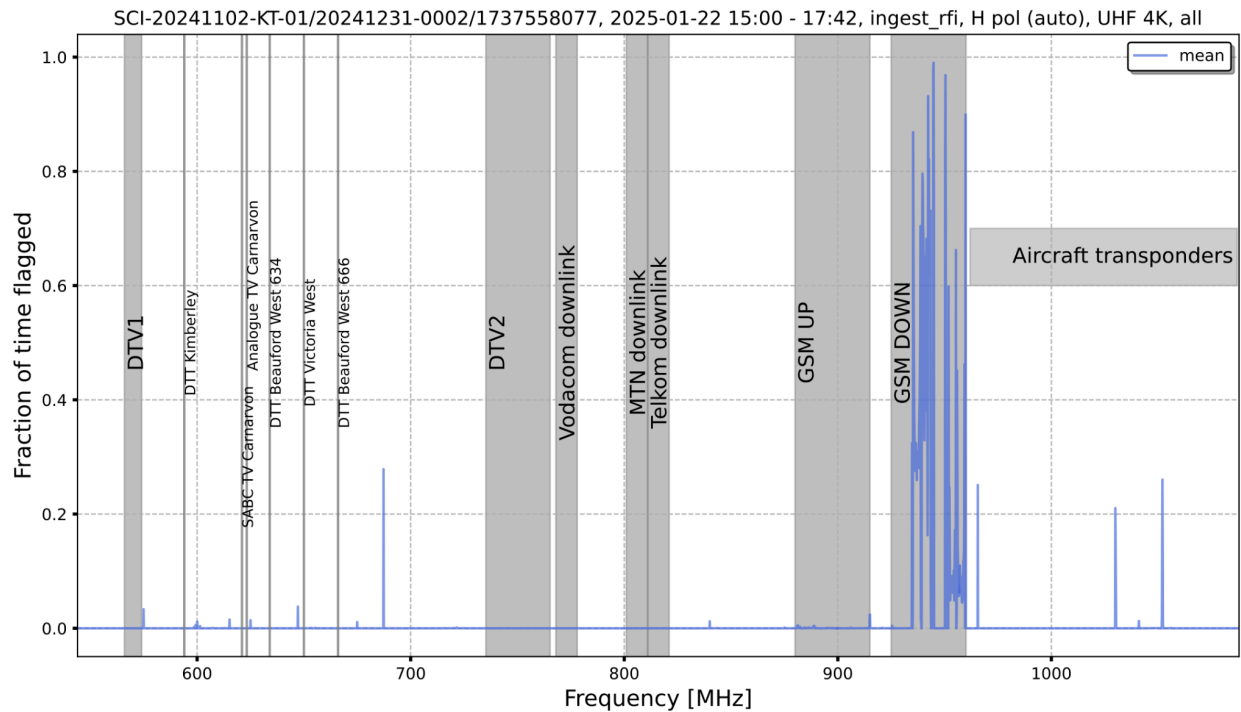


Figure 16: Mean of the autocorrelation flags of SB_ID:20241231-0002 dataset.

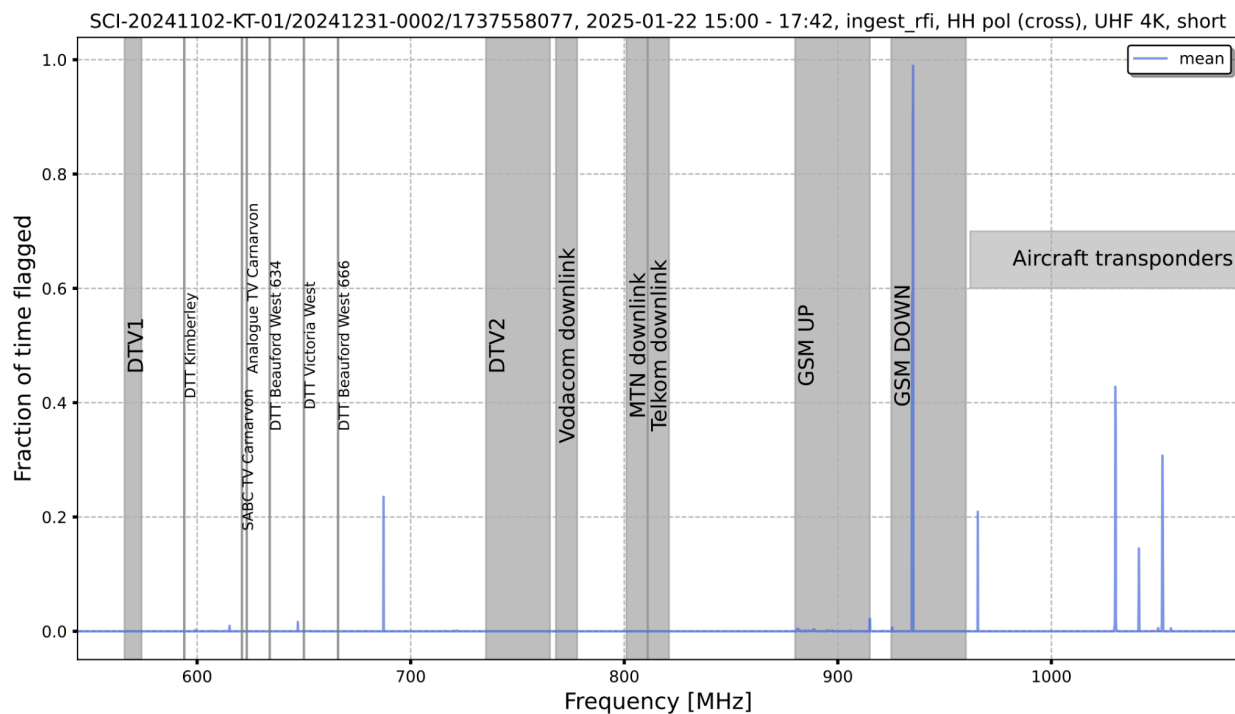


Figure 17: Mean of the cross correlation flags of short baselines of SB_ID:20241231-0002 dataset.

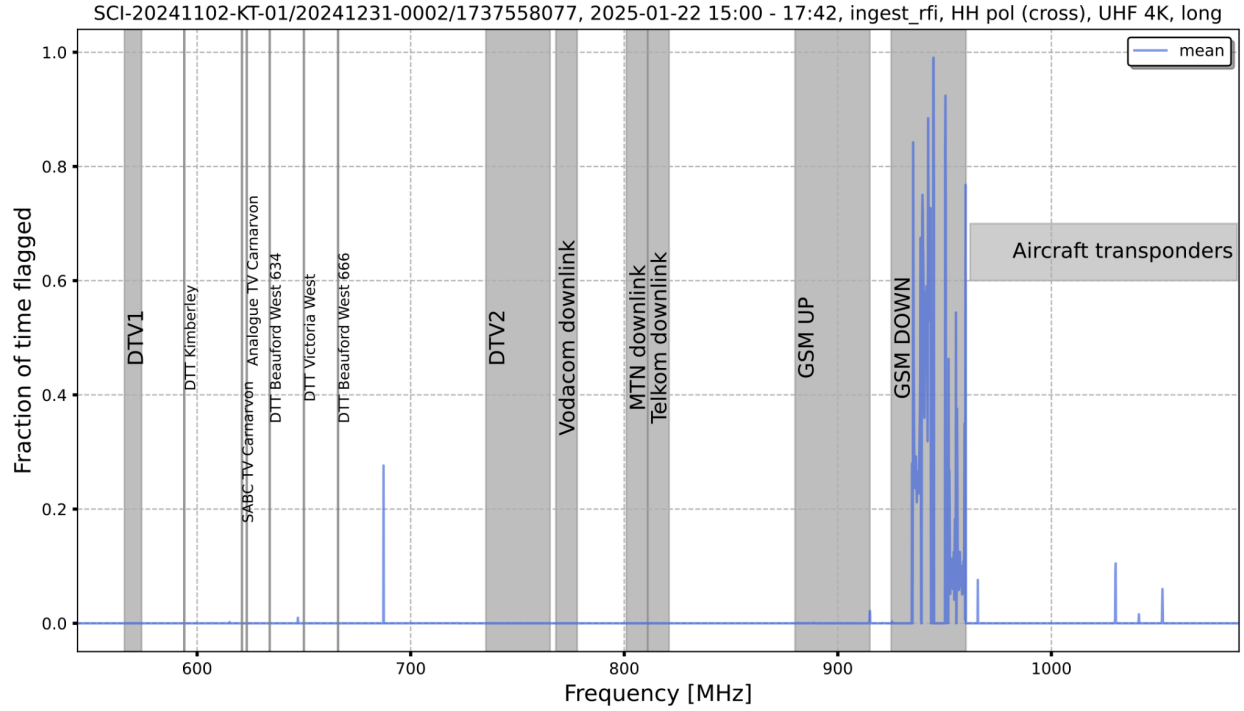


Figure 18: Mean of the cross correlation flags of long baselines of SB_ID:20241231-0002 dataset.

g. 2025, 25 January

Both auto and short baseline cross correlations of the dataset, obtained on the 25th of January 2025, exhibit some channels contaminated by relatively weak RFI emissions, mainly in the lower part of the band (see Figures, 19 and 20). The long baseline cross correlation appears to be clean overall (see Figure 21).

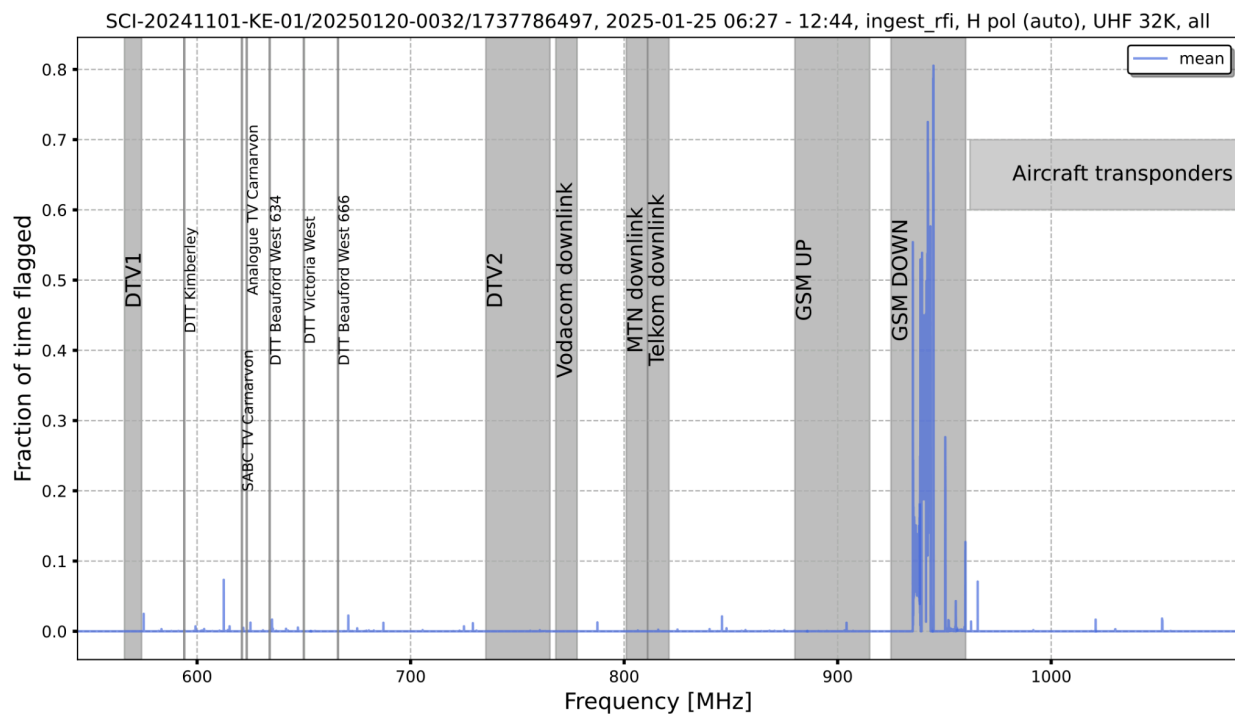


Figure 19: Mean of the autocorrelation flags of SB_ID:20250120-0032 dataset.

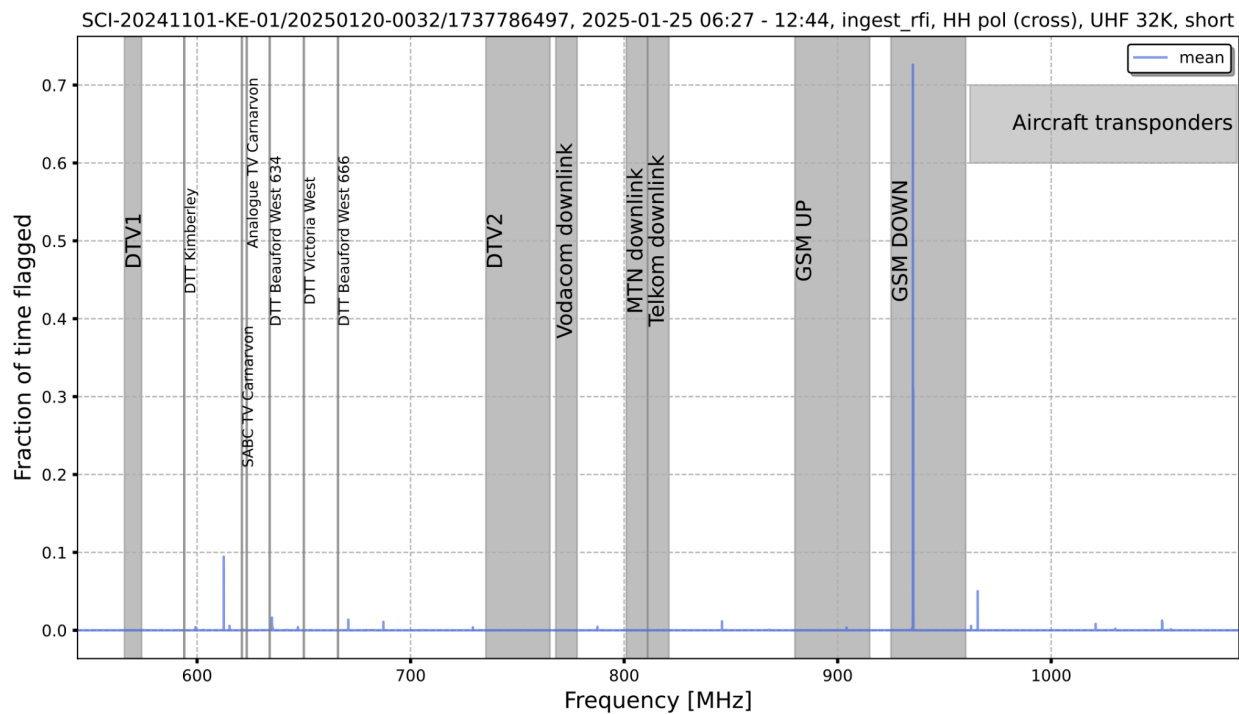


Figure 20: Mean of the cross correlation flags of short baselines of SB_ID:20250120-0032 dataset.

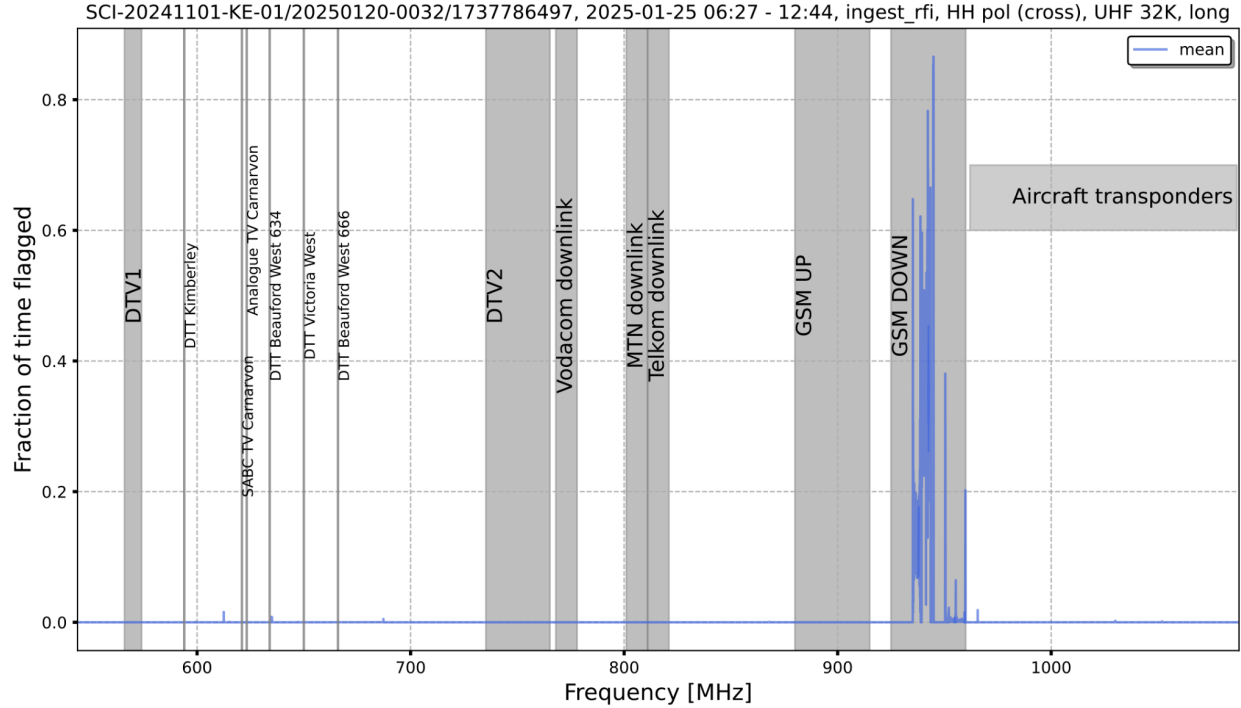


Figure 21: Mean of the cross correlation flags of long baselines of SB_ID:20250120-0032 dataset.

4 . RFI summary for January 2025 at UHF, HH cross polarization

Now we look at the overall RFI for the month of September by concatenating all science observations for the month. Here we select particularly 8s observations, observed at 4k or 32k channel mode. The monitoring and analysis of RFI are presented using the KATHPRFI pipeline [1], highlighting the impact of construction on RFI levels.

KATHPRFI algorithm

The kathprfi essentially processes two numpy arrays, Master and Counter, to update the number of RFI points and observations per voxel which refers to 5-D array of Time, Frequency, Baseline, Elevation and Azimuth. It iterates over chunks of frequency indices, baseline indices, and time indices. For each combination, the algorithm updates the Master array by the corresponding 'Good flags' value and the Counter array by 1. Good flags for a given observation are based on a flag type, e.g. if flag type is calibration RFI then target tags for that

observation will be obtained from a complete calibration report. The output of this is an xarray dataset that contains arrays of these two arrays and the dimensions of interest as mentioned above.

Computation of marginal probability

The RFI probability is computed by using two arrays from the output Zarr file obtained from the kathprfi pipeline : 'Master' and 'Counter'. These arrays represent the number of RFI points and the total number of observations, respectively. The function sums these arrays along a specified dimension to get total counts and then calculates the probability as the ratio of these sums.

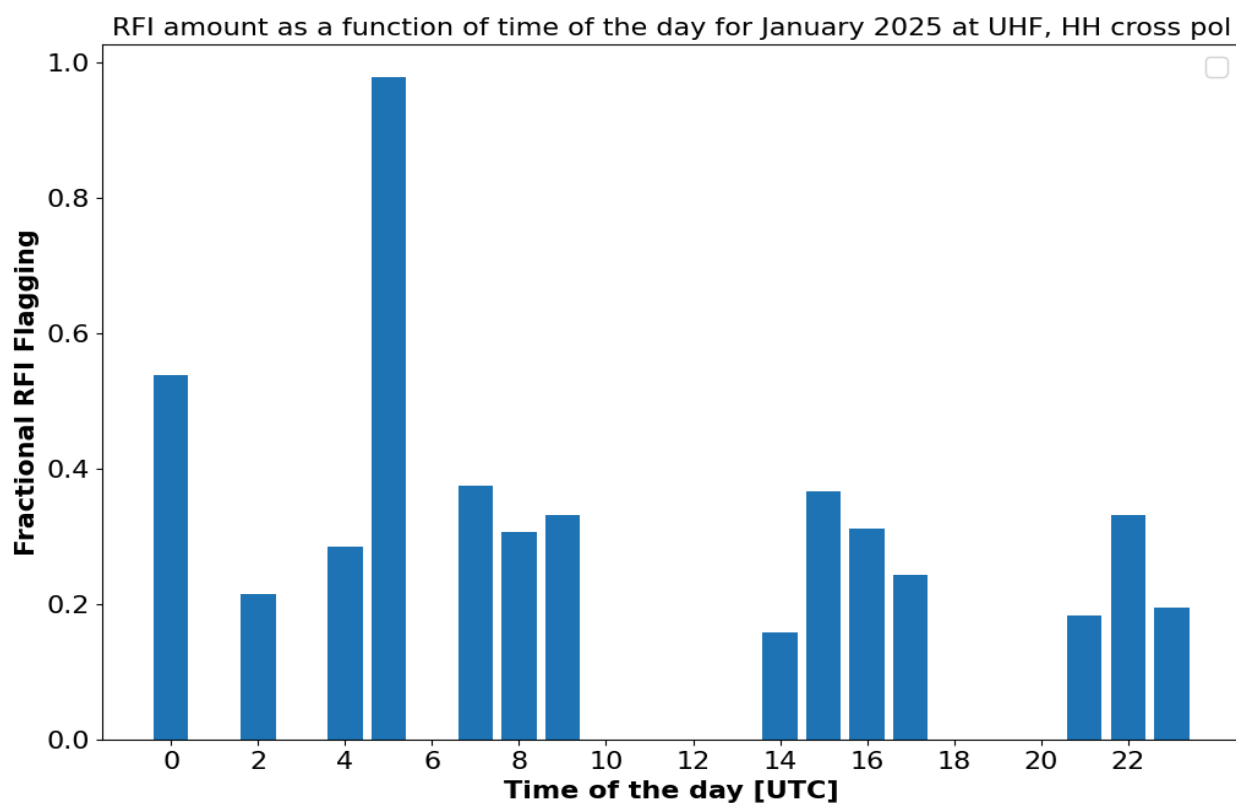


Figure 22: The plot above shows the fractional RFI flagging as a function of time of the day in UTC for January 2025 UHF horizontal cross pol observations.

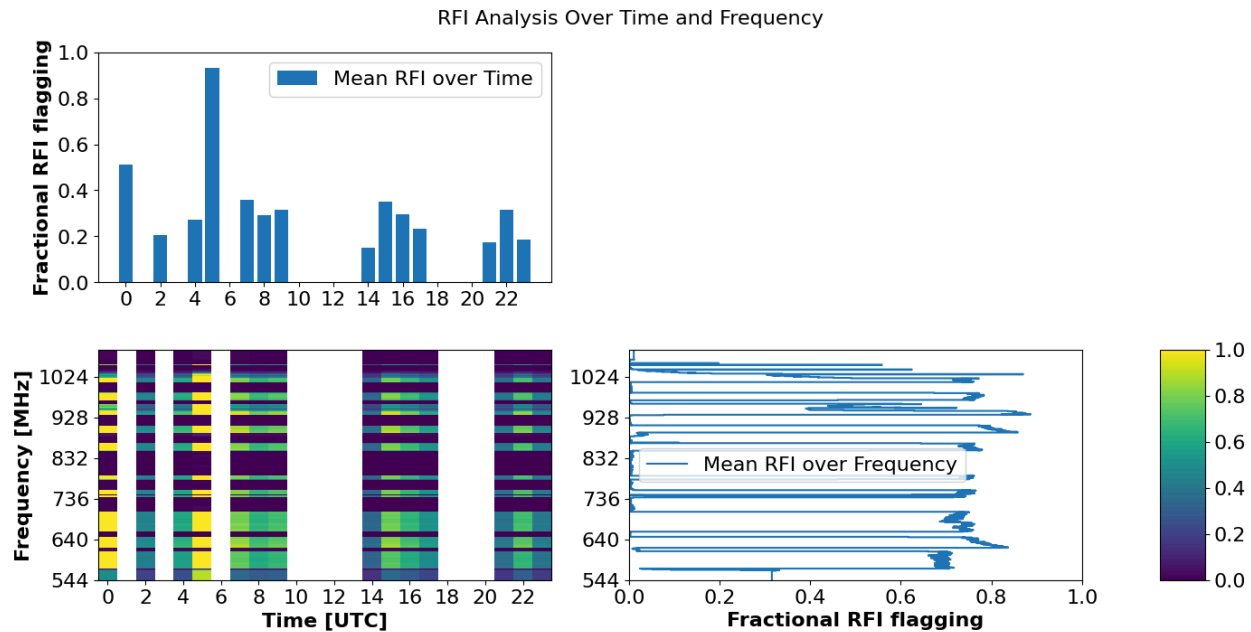


Figure 23: 2-D average RFI occupancy across time and frequency space

Figure 22 illustrates the hourly behavior of RFI for January 2025, highlighting specific time periods affected by elevated RFI levels. The data reveals that RFI behavior is irregularly unstable with time of the day, with an average RFI percentage of 35% throughout the day. Notably, RFI levels exceed the daily average during the hours 00:00 UTC and 05:00 UTC. Among these, a particularly high RFI fraction is observed at (00:00 UTC), where the RFI percentage peaks at above 90%. This suggests a period of heightened RFI prevalence. The causes for such elevated RFI levels may include specific construction activities on-site or interference related to a source under observation during that time. To better characterize this anomalies, a detailed analysis of the frequency spectra during these hours is necessary.

Figure 23 presents a two-dimensional view of RFI patterns across time and frequency. The top 1D bar plot displays mean RFI levels over time, consistent with the trends shown in Figure 22. The central 2D heatmap provides a detailed visualization of RFI activity, with white regions indicating missing data or non-monitored frequency intervals. Intense RFI zones are represented by bright yellow or green patches, as per the color bar on the far right. Significant RFI is observed in specific frequency bands, including 544–592 MHz, 620–650 MHz, 670–700 MHz, 740–760 MHz, 850–870 MHz, 880–900 MHz, 930–950 MHz, 960–990 MHz and above 1000 MHz. The 1D plot on the right highlights average RFI levels across frequencies, where noticeable spikes represent strong interference at specific frequency ranges. Intense RFI is notable for the hours (00:00 UTC, 05:00 UTC) corresponding to the above mentioned frequency intervals.